

Executive Summary

ShopEasy Marketing Analysis | End-to-End Analytics Portfolio Project

1. Company Overview

ShopEasy is an online retail business experiencing declining customer engagement and conversion despite increased investment in digital marketing campaigns. The business needed a data-driven approach to understand customer interactions, identify conversion bottlenecks, evaluate marketing content performance, and uncover customer experience issues from reviews and feedback.

This project delivers an end-to-end marketing analytics solution that transforms raw customer, product, engagement, customer journey, and review data into actionable business insights using SQL, Python, and Power BI.

2. Business Problem

ShopEasy's key challenges were:

- Declining customer engagement with marketing content.
- Lower website conversion and customer progression through the purchase journey.
- Increasing marketing investment without the expected return.
- Limited visibility into customer sentiment and recurring product/service issues.
- A need to identify which products, content types, journey actions, and customer feedback patterns require management attention.

The analysis was designed around the business goals of improving conversion, increasing engagement, and strengthening customer feedback scores.

3. Analytical Approach

The project follows a structured end-to-end analytics workflow:

Business Case → Business Questions & KPIs → Data Preparation → SQL Analysis → Python Sentiment Analysis → Data Modeling → Power BI & DAX → Interactive Dashboard → Insights → Recommendations

SQL Data Preparation & Analysis

SQL was used to prepare and transform the underlying data across products, customers, geography, customer reviews, engagement data, and customer journey tables.

Key preparation activities included:

- Product price categorization into Low, Medium, and High segments.
- Enrichment of customer data using customer-geography relationships.
- Review-text cleaning.
- Standardization of engagement content types.
- Separation of combined views and clicks into individual analytical fields.
- Exclusion of newsletter records from engagement analysis.

- Identification and removal of duplicate customer-journey records using ROW_NUMBER().
- Treatment of missing journey duration using date-level average duration.
- Standardization of journey-stage values.

The business analysis then examined product performance, customer journeys, engagement, ratings, review volume, sentiment, and conversion-related patterns.

Python Customer Sentiment Analysis

Python was used to extend the analysis beyond numerical ratings by applying VADER sentiment analysis to customer review text.

The process:

- Extracted customer review data directly from SQL Server.
- Calculated a normalized VADER compound sentiment score from -1 to +1.
- Combined textual sentiment with numerical ratings.
- Classified reviews into Positive, Mixed Positive, Neutral, Mixed Negative, and Negative categories.
- Created sentiment-score buckets.
- Exported the enriched review dataset for Power BI analysis.

Power BI & DAX

The final Power BI solution brings the analytical layers together through interactive reporting, KPI cards, time-series analysis, funnel analysis, product-level analysis, sentiment analysis, slicers, conditional formatting, and MoM performance indicators.

The dashboard enables management to monitor:

- Total Customers
- Conversion Rate
- Engagement Rate
- Average Rating
- Views
- Clicks
- Likes
- Customer Journey Actions
- Product-level conversion
- Customer review and sentiment performance
- Month-over-Month KPI movement

4. Key Insights

1. The conversion funnel is a critical area for optimization

The dashboard's dedicated Conversion Details analysis evaluates customer progression through journey actions and stages rather than relying only on overall conversion. This provides visibility into where customers are engaging with the buying journey and where drop-offs occur.

The analysis specifically incorporates customer journey actions, including explicit Drop-off activity, making the funnel diagnosis more actionable than simply comparing stage volumes.

Business implication: Improving conversion requires attention to the customer journey itself—not simply generating additional traffic.

2. Engagement should be evaluated beyond views

The dashboard tracks Views, Clicks, and Likes separately and combines them into an overall Engagement Rate. This helps distinguish between content that generates visibility and content that encourages interaction.

The engagement analysis also compares performance across content types and over time, directly addressing the business case objective of determining which content formats generate stronger customer interaction.

Business implication: Marketing decisions should prioritize content that generates meaningful interaction rather than optimizing purely for reach.

3. Product performance and customer sentiment need to be considered together

The dashboard connects product-level review analysis with average ratings, review volume, and sentiment categories. This creates an important distinction between:

- Products with strong customer perception.
- Products generating substantial customer feedback.
- Products receiving weaker ratings.
- Products where negative sentiment may require investigation.

The Python sentiment layer strengthens this analysis by combining review text sentiment with numerical ratings, rather than treating ratings alone as the complete picture of customer experience.

Business implication: Products receiving recurring negative feedback should be investigated for underlying quality, value, delivery, or customer-experience issues.

4. Customer feedback provides an additional diagnostic layer

The customer-review dashboard provides visibility into:

- Rating distribution.
- Average rating.
- Review volume.
- Sentiment-category distribution.
- Product-level reviews.
- Review text alongside sentiment classification.

This directly supports ShopEasy's requirement to identify recurring positive and negative customer feedback themes.

Business implication: Customer feedback should be treated as an input into product improvement and marketing strategy rather than simply as a reporting metric.

5. MoM tracking makes the dashboard performance-oriented

The dashboard incorporates Month-over-Month percentage analysis and directional arrows for key KPIs such as:

- Customer count
- Engagement Rate
- Conversion Rate
- Average Rating
- Views
- Clicks
- Likes

This allows stakeholders to quickly identify whether performance is improving or deteriorating rather than looking only at absolute KPI values.

Business implication: Management can use the dashboard as an ongoing performance-monitoring tool instead of a static analytical report.

5. Recommendations

1. Prioritize lower-funnel conversion optimization

Investigate the customer journey actions and stages associated with the highest drop-off activity.

Focus on potential friction points such as:

- Product-page experience.
- Checkout usability.
- Purchase barriers.
- Calls-to-action.
- Product information.
- Customer trust factors.

The business case explicitly identifies funnel drop-off analysis as a key requirement for improving conversion.

2. Allocate marketing effort based on engagement quality

Use Engagement Rate, Views, Clicks, and Likes together when evaluating marketing content.

Content generating high reach but weak interaction should be reviewed, while consistently engaging content should receive greater consideration for future campaigns.

3. Improve campaign efficiency through KPI monitoring

Track Conversion Rate and Engagement Rate alongside their MoM movement to determine whether increased marketing activity is translating into meaningful customer actions.

Marketing investment should ultimately be evaluated against customer response and conversion outcomes rather than activity volume alone.

4. Act on negative customer sentiment

Create a recurring review process for products with:

- Low average ratings.
- High negative-review volume.
- Negative sentiment patterns.
- High review activity combined with poor customer perception.

These products should be prioritized for root-cause investigation and corrective action.

5. Close the loop between customer feedback and product strategy

Use sentiment and review analysis to identify recurring customer concerns and feed those findings back into:

Product → Marketing → Customer Experience → Conversion

Improving the customer experience can strengthen satisfaction while also supporting future engagement and conversion.

6. Use the dashboard for continuous performance monitoring

Establish a regular KPI review cadence covering:

Customers → Engagement → Conversion → Ratings → Sentiment → Customer Journey

The MoM indicators make it easier to identify emerging changes and determine where corrective action is required.

6. Business Impact

This project demonstrates how an organization can move from fragmented marketing, customer journey, and feedback data to a unified decision-support solution.

- The final analytical workflow enables ShopEasy to:
- Identify customer journey drop-off points.
- Monitor conversion performance.
- Evaluate marketing engagement.
- Compare content performance.
- Track customer and engagement trends.
- Analyze product-level customer feedback.
- Combine numerical ratings with text sentiment.
- Identify products requiring customer-experience attention.
- Monitor KPI performance through MoM analysis.
- Translate analytical findings into targeted marketing and customer-experience recommendations.

Overall, the project addresses the core challenge identified in the business case: understanding why increased marketing activity is not translating into stronger engagement and conversion, and using customer feedback to identify opportunities for improvement.

Executive Takeaway

The ShopEasy Marketing Analysis demonstrates a complete analytics lifecycle—from business problem definition and SQL data preparation to Python-based sentiment analysis, data modeling, DAX-driven KPI development, and interactive Power BI reporting. The analysis provides management with a unified view of customer engagement, conversion performance, journey behavior, product feedback, and sentiment, enabling more targeted marketing decisions and customer-experience improvements.

The primary strategic opportunity is to shift from simply measuring marketing activity toward optimizing the complete customer journey: attract the right audience, create meaningful engagement, reduce funnel drop-offs, understand customer feedback, and continuously monitor KPI movement to drive sustainable conversion improvement.